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BANK OF CHINA (HONG KONG)



中銀香港極客大賽 BOCHK FinTech Hackathon

IBM Cloud Services/APIs

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Cloudant (No-SQL database)	Data

Watson Conversation Service

- ▶ Add a natural language interface to your application to automate interactions with your end users. Common applications include virtual agents and chat bots that can integrate and communicate on any channel or device. Train Watson Conversation service through an easy-to-use web application, designed so you can quickly build natural conversation flows between your apps and users, and deploy scalable, cost effective solutions.

The screenshot shows the Watson Conversation Service interface. On the left, there is a list of intents under the heading "Car Dashboard Sample". The list includes:

- #book_room: I need a room for this weekend
- #no: Absolutely not
- #yes: absolutely
- #locate_amenity: Does this hotel have a bar?

The right side of the interface shows a "Try it out" dialog flow. The flow starts with the user saying "Hey Watson, I need a room". The system responds with "We've only got one room left - would you like it?". The user says "nah". The system responds with "Fair enough!". The user asks "Where's the bar?". The system responds with "The bar is located down the corridor to your right".

The screenshot shows the Watson Conversation Service interface for a "Booking conversation bot". The flow starts with the user saying "Which amenity do you need?". The system responds with "Watson says". The user says "Anything else". The system responds with "This 'Anything else' node enables you to provide a fallback answer when Watson doesn't understand the user's intent."

A dialog uses the intent and entity that have been identified, plus context from the APP, to interact with the user and ultimately provide a response. The response might be the answer to a question such as "Where is the pool?" or the execution of a command, such as turning on the radio. The intent and entity might be enough information to identify the correct response, or the dialog might ask the user for more input that is needed to respond correctly.

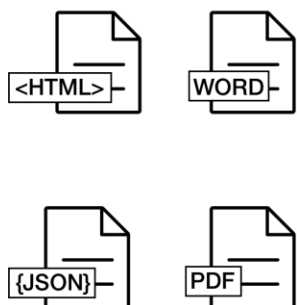
[View dialog tutorial](#)

Watson Discovery Service

- ▶ Add a cognitive search and content analytics engine to applications to identify patterns, trends and actionable insights that drive better decision-making. Securely unify structured and unstructured data with pre-enriched content, and use a simplified query language to eliminate the need for manual filtering of results.

| Data

Private data



| Ingestion

Convert and enrich by leveraging Watson APIs to add NLP meta data to your content, making it easier to explore and discover insights

Clean and normalize through an automated processing of NLP results, improving data quality

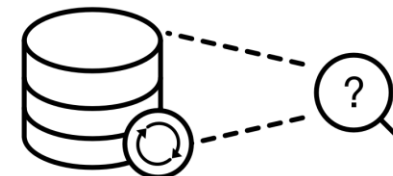
| Storage

Normalized data is **indexed into a collection** as part of your **environment** in the cloud



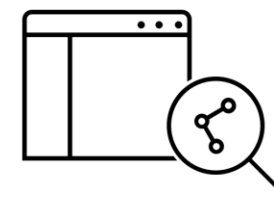
| Query

Understand data faster, create better hypothesis and deliver better outcomes



| Output

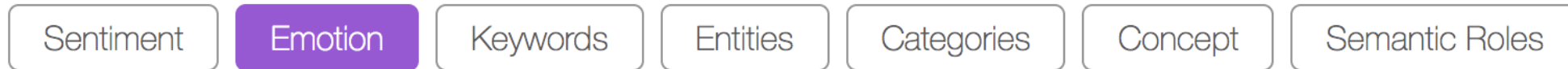
Actionable insights into your app





Natural Language Understanding

- Analyze text to extract meta-data from content such as concepts, entities, keywords, categories, sentiment, emotion, relations, semantic roles, using natural language understanding. With custom annotation models developed using Watson Knowledge Studio, identify industry/domain specific entities and relations in unstructured text.



Analyze the overall emotion and the targeted emotion of the content.

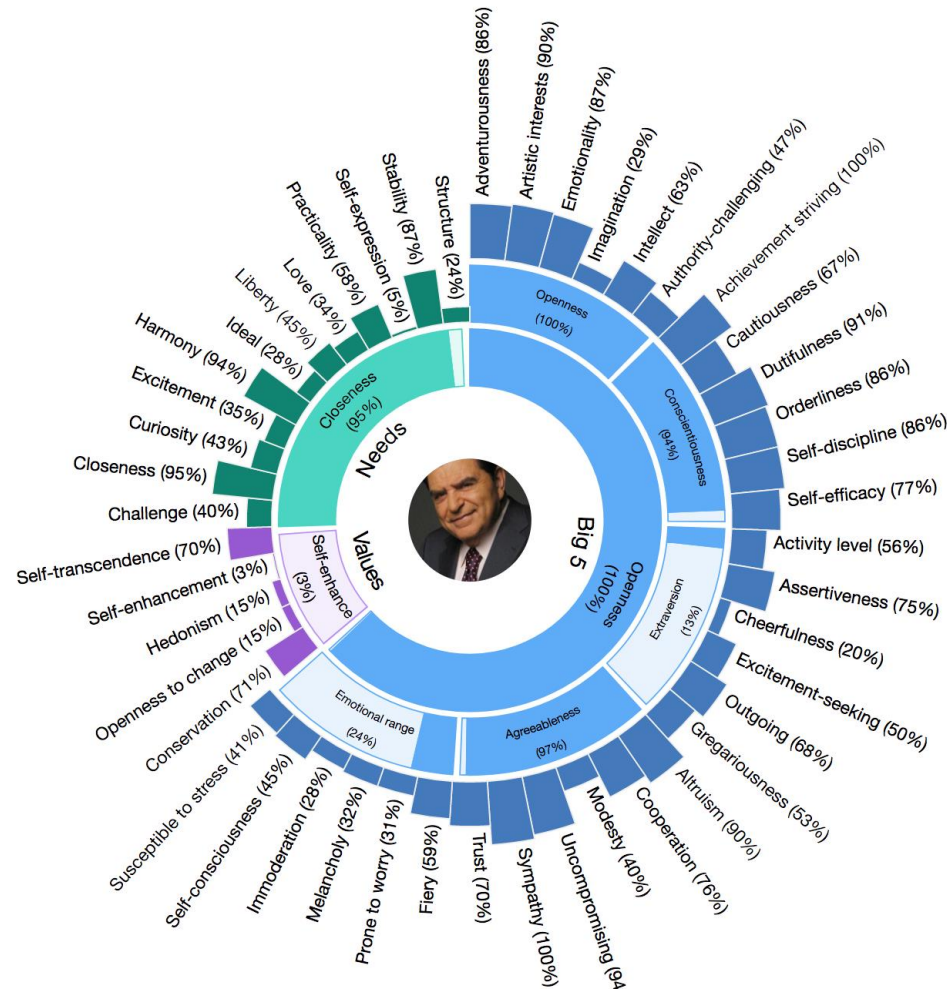
[JSON](#) ✓

Overall Emotion



Personality Insights

- ▶ Personality Insights derives insights from transactional and social media data to identify psychological traits which determine purchase decisions, intent and behavioral traits; utilized to improve conversion rates.



Tone Analyzer

- Cognitive linguistic analysis to identify a variety of tones at both the sentence and document level. This insight can then be used to refine and improve communications. It detects three types of tones, including emotion (anger, disgust, fear, joy and sadness), social propensities (openness, conscientiousness, extroversion, agreeableness, and emotional range), and language styles (analytical, confident and tentative) from text.

Identify sentences with stronger tones in context or sorted by score. Highlighted sentences indicate the likelihood of a tone present. If more than one tone is present, the stronger one is shown. Click on a sentence to see a breakdown of all tones.

Tones

- Analytical
- Anger**
- Confident
- Fear
- Tentative

In context

Anger: Evoked due to injustice, conflict, humiliation, negligence or betrayal. If anger is active, the individual attacks the target, verbally or physically. If anger is passive, the person silently sulks and feels tension and hostility.

< .5

.5 - .75

> .75

NoneStrong

I hate these new features On #ThisPhone after the update.

I hate #ThisPhoneCompany products, you'd have to torture me to get me to use #ThisPhone.

The emojis in #ThisPhone are stupid.

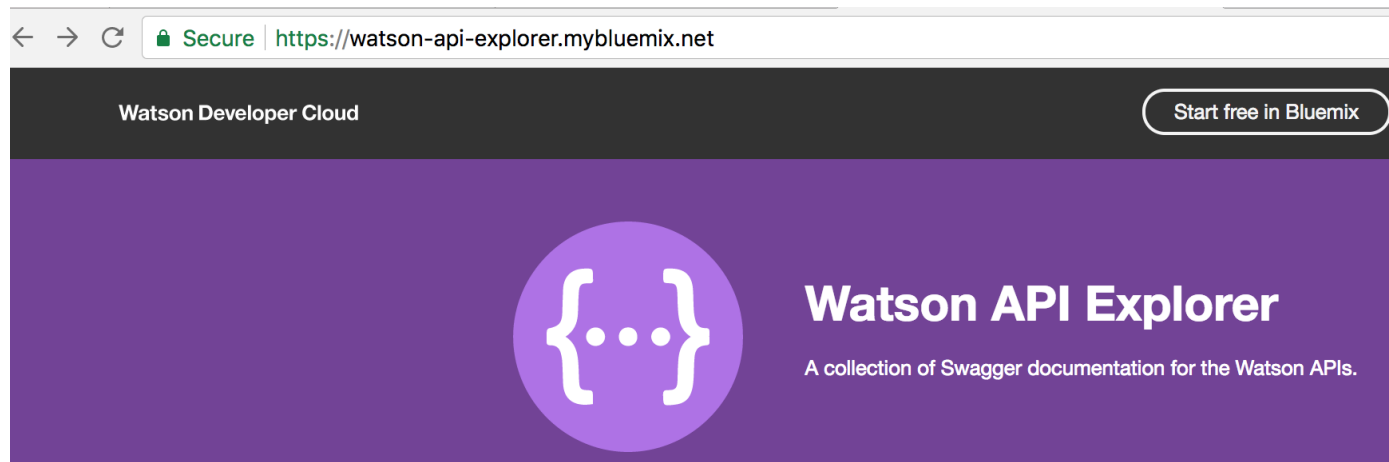
#ThisPhone is a useless, stupid waste of money.

#ThisPhone is the worst phone I've ever had - ever 😞.

#ThisPhone another ripoff, lost all respect SHAME.

I'm worried my #ThisPhone is going to overheat like my brother's did.

Watson Developer Portal



Watson API Explorer

Conversation

The IBM Watson™ Conversation service combines machine learning, natural language understanding, and integrated dialog tools to create conversation flows between your apps and your users.

For more information about this service, see the documentation:

<https://console.bluemix.net/docs/services/conversation/index.html>

Workspaces

Show/Hide | List Operations | Expand Operations

GET	/v1/workspaces	List workspaces
POST	/v1/workspaces	Create workspace
DELETE	/v1/workspaces/{workspace_id}	Delete workspace
GET	/v1/workspaces/{workspace_id}	Get information about a workspace
POST	/v1/workspaces/{workspace_id}	Update workspace

Message

Show/Hide | List Operations | Expand Operations

POST	/v1/workspaces/{workspace_id}/message	Get a response to a user's input
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Watson API Explorer

Visual Recognition

Important: As of September 8, 2017, the beta period for Similarity Search is closed. For more information, see [Visual Recognition API - Similarity Search Update](#).

The IBM Watson™ Visual Recognition service uses deep learning algorithms to identify scenes, objects, and faces in images you upload to the service. You can create and train a custom classifier to identify subjects that suit your needs.

Tip: To test calls to the **Custom classifiers** methods with the API explorer, provide your **api_key** from your IBM® Cloud service instance.

For more information about this service, see the API reference:

<http://www.ibm.com/watson/developercloud/visual-recognition/api/v3/>

General

Show/Hide | List Operations | Expand Operations

GET	/v3/classify	Classify an image
POST	/v3/classify	Classify images

Face

Show/Hide | List Operations | Expand Operations

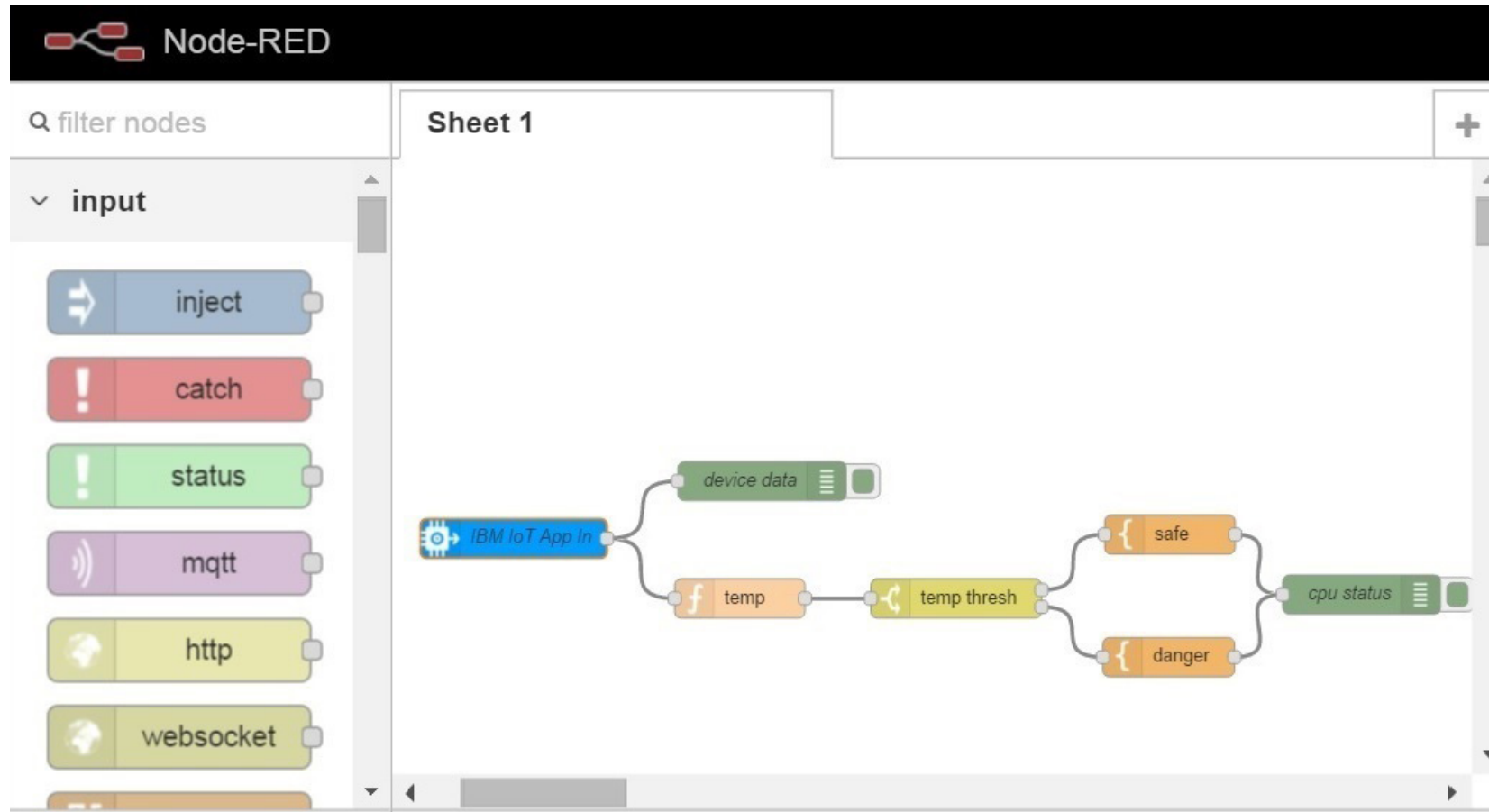
GET	/v3/detect_faces	Detect faces in an image
POST	/v3/detect_faces	Detect faces in images

Custom

Show/Hide | List Operations | Expand Operations

Node-Red

- ▶ Node-RED provides a browser-based flow editor that makes it easy to wire together devices, APIs, and online services by using the wide range of nodes in the palette. Flows can be then deployed to the Node.js runtime with a single click.



Data Science Experience

- Master the art of data science with Data Science Experience (DSX). Solve your toughest data challenges with the best tools and the latest expertise in a social environment built by data scientists.

The screenshot displays the Data Science Experience (DSX) interface, which is a collaborative data science environment. It is divided into two main panels, each showing a different project.

Left Panel: Automobiles - Insights from Twitter data about...

- Code Editor:** Shows a Jupyter Notebook cell with the following code:

```
p_tweets_grouped_by_sentiment = None
```
- Article Preview:** A sidebar on the right shows a preview of an article titled "Interactive time series with dygraphs" from the RStudio Blog, dated April 14, 2015. The article is categorized under "Visualization".
- Text:** Below the code, there is a section titled "Analyze timelines" with the text: "To learn more about which car manufacturing events occurred in 2015, you can plot data following section group all tweets created in 2015 by their posting date (and their sentiment tweets per date)." Below this is a code cell with the following code:

```
In [18]: df_num_tweets_and_sentiment_over_time = df_cleaned_tweets.groupby('POSTING_TIME').agg(F.count('MESSAGE_BODY').alias('NUM_TWEETS')).order_by('POSTING_TIME', ascending=True)df_num_tweets_over_time = df_num_tweets_and_sentiment_over_time.groupby('POSTING_TIME').agg(F.sum('NUM_TWEETS').alias('NUM_TWEETS')).order_by('POSTING_TIME', ascending=True)p_df_num_tweets_and_sentiment_over_time = df_num_tweets_and_sentiment_over_time.toPandas()p_df_num_tweets_over_time = df_num_tweets_over_time.toPandas()
```
- Footer:** A message at the bottom says "Run the next cell to prepare the data for plotting by moving data with different sentiment".

Right Panel: Car Analysis - Visualize car data with Brunel

- Code Editor:** Shows a Jupyter Notebook cell with the following code:

```
In [6]: %brunel data('cars') x(mpg) y(horsepower) color(origin) :: width=1000, height=400
```
- Table:** A table of car data is displayed above the plot:

	mpg	displacement	horsepower	weight	acceleration	year	origin	name
3	16.0	8	304.0	150.0	3433	12.0	American	amc rebel sst
4	17.0	8	302.0	140.0	3449	10.5	American	ford torino
5	15.0	8	429.0	198.0	4341	10.0	American	ford galaxie 500
- Figure:** A scatter plot titled "Visualize car data with Brunel" showing Horsepower (Y-axis, 60 to 220) versus Mpg (X-axis, 10 to 45). The data points are colored by origin: American (blue), European (red), and Japanese (yellow). The plot shows a clear negative correlation between horsepower and miles per gallon.
- Files Panel:** A sidebar on the right shows a list of files: "dac_sample.txt", "precipitation.csv", "NYPD_Motor_Vehicle_Collisions.csv", and "cars.csv". Each file has an "Insert to code" button.

Push Notification

- ▶ Push Notifications service provides a unified push service to send real-time notifications to mobile and web applications. The service provides the ability to personalize and send notifications to a segment of users, single user or broadcast to all users.

Push Notifications - Choose recipients

The screenshot displays a web interface for sending push notifications. On the left, a sidebar contains four menu items: 'Send Notifications' (highlighted in blue), 'Manage Tags', 'Configure', and 'Monitoring'. The main content area is titled 'Send Notifications' and is divided into two sections: 'Compose' and 'Optional Settings'.

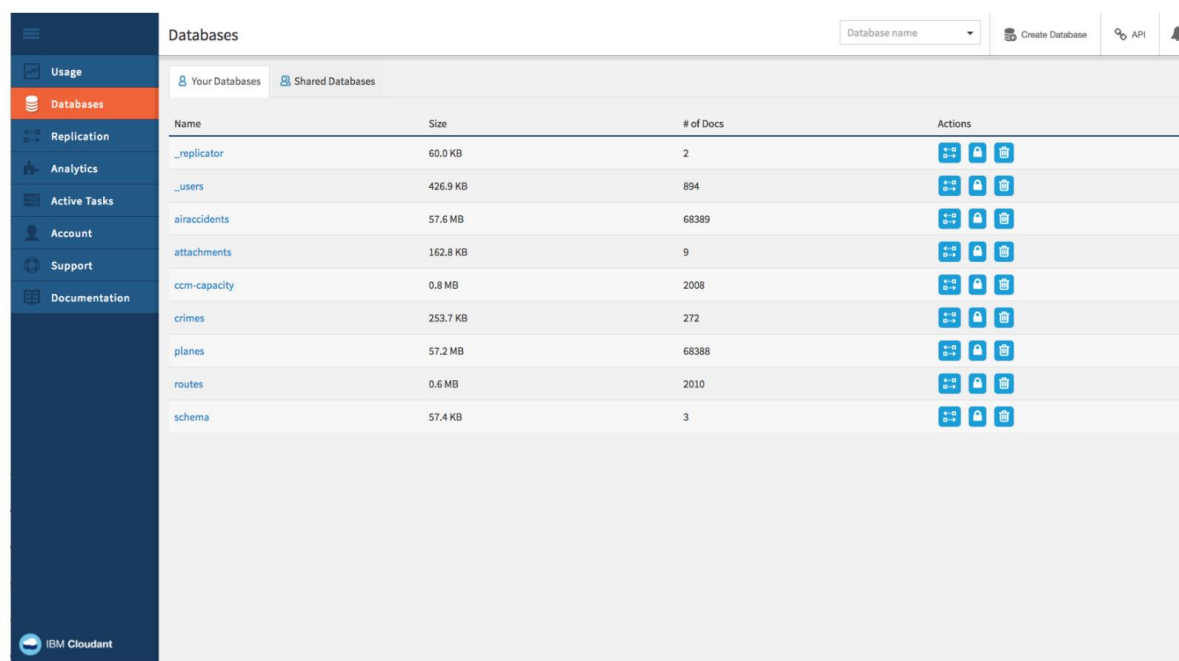
Compose Section:

- Send To:** A dropdown menu currently set to 'All Devices'.
- Message:** A large text input area with the placeholder text 'Type your message here'.
- Additional Payload:** A text input area containing the JSON payload `{"key": "value"}`.

Optional Settings Section:

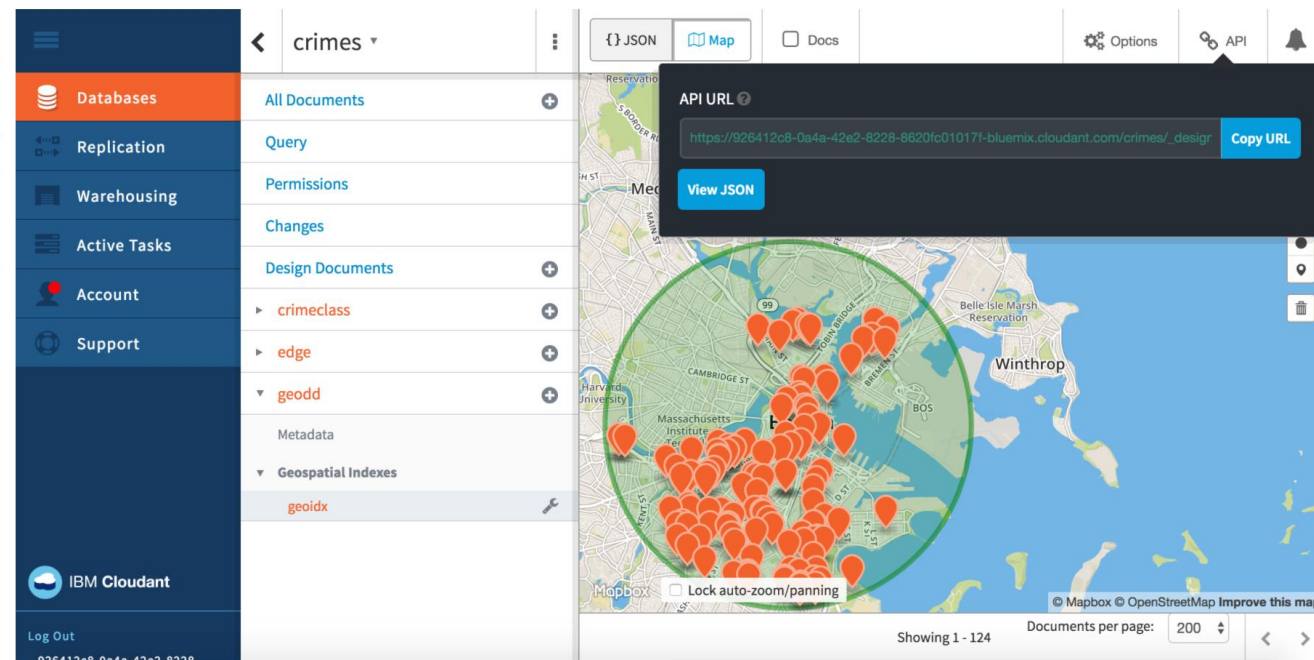
- Platform Selection:** Two buttons, 'Mobile' (active) and 'Web'.
- Device Types:** Two expandable sections, 'iOS' and 'Android', each preceded by a right-pointing chevron (>).

- ▶ Cloudant NoSQL DB provides access to a fully managed NoSQL JSON data layer that's always on. This service is compatible with CouchDB, and accessible through a simple to use HTTP interface for mobile and web application models



The screenshot shows the 'Databases' section of the Cloudant console. It features a table with columns for Name, Size, # of Docs, and Actions. The table lists several databases, including system databases like _replicator and _users, and user databases like airaccidents, attachments, ccm-capacity, crimes, planes, routes, and schema. Each database has a set of action icons (plus, lock, and trash) in the Actions column.

Name	Size	# of Docs	Actions
_replicator	60.0 KB	2	[+][lock][trash]
_users	426.9 KB	894	[+][lock][trash]
airaccidents	57.6 MB	68389	[+][lock][trash]
attachments	162.8 KB	9	[+][lock][trash]
ccm-capacity	0.8 MB	2008	[+][lock][trash]
crimes	253.7 KB	272	[+][lock][trash]
planes	57.2 MB	68388	[+][lock][trash]
routes	0.6 MB	2010	[+][lock][trash]
schema	57.4 KB	3	[+][lock][trash]



The screenshot shows the 'crimes' database interface in Cloudant. The left sidebar contains navigation links for Databases, Replication, Warehousing, Active Tasks, Account, and Support. The main content area shows a list of actions for the 'crimes' database: All Documents, Query, Permissions, Changes, Design Documents, crimeclass, edge, geodd, Metadata, and Geospatial Indexes. The 'geodd' index is selected, and a map view is displayed. The map shows a geographic area with numerous orange location pins. A modal window is open over the map, displaying the API URL: https://926412c8-0a4a-42e2-8228-8620fc010171-bluemix.cloudant.com/crimes/_design/. The modal also includes a 'Copy URL' button and a 'View JSON' button. The bottom of the interface shows 'Showing 1 - 124' and 'Documents per page: 200'.

IBM Cloud (PaaS)

